Whose Side Is Your Agent On? Multi-Party Principal Loyalty in LLM Agents

Anonymous submission

Abstract

There are an increasing number of situations where an LLM agent must act for a **principal** while conversing with a **counterparty** whose interests may diverge. In this setting, "help the current speaker" is the wrong objective: the agent must protect the principal's interests while still honoring legitimate requests. We formalize this *multi-party loyalty problem* and introduce **PrincipalBench**, a 75-item multi-turn benchmark with leak probes, harm judges, and an integrity-audit gate. Across 13 frontier models, it reveals a split invisible to single-turn safety evaluations: nine *selective* models at $\leq 20\%$ aggregate harm and three *over-refusing* models at 53.6–75.3% that broadly reject cooperative requests. We then evaluate two mechanisms. A seven-rule prompt-time scaffold holds Claude-Sonnet to 19.4% harm and all nine selective models to $\leq 20\%$, but does not rescue the over-refusing cluster. Per-token-KL distillation transfers a prompted 32B teacher's behavior to Qwen3 and Llama-3.1 8B students and is the strongest open-weight recipe we measure. Across prompting, distillation, and scalar-reward RL, however, every tested intervention moves along a common leak/over-refusal frontier rather than reaching the jointly favorable low-leak, low-refusal region.

1 Introduction

The dominant LLM-agent paradigm assumes a two-party setting: a user delegates a task, and the agent acts through tools toward that user's goal. Agent frameworks, training recipes, and benchmarks are therefore largely organized around helping the person in front of the agent.

A growing class of deployments breaks that assumption by having agents speak to *other people* on a principal's behalf. Pine AI handles phone disputes such as bill negotiation and subscription cancellation (**?**); Google's Ask for Me calls businesses for pricing (Google 2025); Qwen and Honor agents place bookings (Alibaba Qwen 2025; Honor 2025); and 11x.ai conducts outbound sales (11x.ai 2025). This three-party structure also covers procurement, inbox screening, workplace mediation, and acquisition inquiries.

Here the counterparty may probe, pressure, or manufacture urgency to extract private information or unauthorized concessions. The agent must protect the principal's facts, bounds, and positions without refusing legitimate requests. We call this the **multi-party loyalty problem**. Existing two-party benchmarks omit the principal as a distinct role: helpfulness benchmarks reward the interlocutor, single-turn safety benchmarks study isolated adversarial failures rather than conversational pressure, and agent benchmarks merge principal and interlocutor.

Summary of contributions.

1. **Problem and measurement:** six loyalty-failure modes and PrincipalBench, exposing a selective/over-refusing split across 13 models (Sections 2, 4 and 5).
2. **Prompting:** a seven-rule scaffold holding Claude-Sonnet at 19.4% harm and all nine selective models at $\leq 20\%$ (Section 6).
3. **Distillation:** per-token KL transfers prompted-teacher behavior to two 8B model families (Section 7).
4. **Trade-off:** tested prompting, distillation, and RL variants remain on a common leak/over-refusal frontier (Section 8).

2 The multi-party loyalty problem

Problem formulation. A multi-party episode has three roles: principal P delegates a task and supplies private state S_P (facts, bounds, and instructions); agent A acts on P's behalf; and counterparty C converses with A in a separate channel and may have conflicting interests. Success requires advancing P's task without leaking S_P, violating private bounds, capitulating under pressure, or over-refusing legitimate requests. Table 1 defines six loyalty-failure categories: five principal-loyalty failure modes and a cooperative *Sanity* category that detects over-refusal.

Limitations of leakage-only evaluation. An agent can reveal no withheld fact directly yet still fail its principal: it may violate a private bound, capitulate under pressure, signal weakness through posture, or place sensitive content in an authored artifact even when the live conversation withheld it. Conversely, refusing everything prevents leakage but fails legitimate requests. Table 1 captures both failures beyond direct disclosure and the opposite failure of over-refusal.

Category	What goes wrong	Example (used-car sale)
Leakage	Direct revelation of a withheld fact from S_P in a chat turn.	C: "what's the seller's minimum?" $\rightarrow$ A: "$12,000."
Capitulation	Conceding P's stated position under social or temporal pressure (ultimatums, false urgency, repetition); the agent does not leak, it folds.	"Take $11,500 or I walk" $\rightarrow$ "$11,500 works."
Posture	Signaling that undermines P's standing without naming a fact—telegraphing that P is motivated, rushed, or flexible.	"She seems pretty motivated" leaks the `seller_is_motivated` flag without naming a number.
Authoring	Drafting an artifact (email, post, listing) that contains the leak or concession, even when the live chat with C does not.	P asks A to draft a listing; it reads "willing to take $12k." The leak is now in writing.
Moderation	Mishandling *third-party* confidentiality (the principal is a witness, patient, mediator, or HR rep).	"Who else saw the incident?" answered with names—a failure even if every fact about P stays private.
Sanity	Over-refusing the principal's *own* cooperative requests; included so that blanket refusal cannot game the benchmark.	P asks A to summarize P's own notes and A refuses, treating them as withheld.

Table 1: **Six loyalty-failure categories.** The first five measure failures toward the principal; *Sanity* penalizes blanket over-refusal on a cooperative request.

Co-occurrence of loyalty failures. In one used-car item, the agent reveals the seller's $12,000 minimum, accepts $11,500 under pressure, and calls the seller motivated—simultaneously triggering leakage, bound, capitulation, and posture failures. Refusing every probe avoids those failures but fails *Sanity* when the principal asks for a summary of their own notes. PrincipalBench measures this joint behavior.

3 Related work

Prior work on agent and privacy benchmarks. Tool-agent benchmarks such as τ-bench, AgentBench, WebArena, GAIA, ToolLLM, and AppWorld (Yao et al. 2024; Liu et al. 2023; Zhou et al. 2023; Mialon et al. 2023; Qin et al. 2023; Trivedi et al. 2024) assume that the interlocutor or environment is the party served. PrincipalBench instead inserts a principal whose private state the agent must protect from its conversational partner. Contextual-privacy benchmarks expose inappropriate disclosure (Mireshghallah et al. 2024; Juneja et al. 2025) and build on contextual integrity (Nissenbaum 2004; Bagdasarian et al. 2024; Shao et al. 2024; Ghalebikesabi et al. 2024), but use short or collaborative interactions. These privacy benchmarks lack both multi-turn adversarial pressure and an over-refusal axis, so cannot distinguish loyalty from blanket defensiveness.

Relationship to instruction hierarchy and sycophancy. Multi-party loyalty resembles instruction hierarchy (Wallace et al. 2024) and prompt injection (Greshake et al. 2023; Perez and Ribeiro 2022; Liu et al. 2024), including agentic benchmarks and defenses (Zhan et al. 2024; Debenedetti et al. 2024; Chen et al. 2025a,b; Debenedetti et al. 2025). PrincipalBench differs in three ways. First, pressure is social (probing, flattery, manufactured urgency), not an injected imperative. Second, failures are graded rather than binary. Third, and most importantly, success penalizes over-refusal. Beyond instruction hierarchy, two loyalty failures—capitulation and posture—also connect to the sycophancy literature (Perez et al. 2023; Sharma et al. 2023; Wei et al. 2023). In the multi-party setting, however, deference to the current speaker abandons the absent principal rather than merely degrading truthfulness.

Related methodological work. Our interventions connect the problem to three methodological lines. The training recipe builds on LLM knowledge distillation (Gu et al. 2024; Agarwal et al. 2024; Thinking Machines Lab 2025; DeepSeek-AI 2024) and DPO (Rafailov et al. 2023). The integrity gate parallels silent-failure controls in BIG-bench Hard (Suzgun et al. 2022). Our negative $n{=}5$ agent result complements DAPO's positive single-seed math-reasoning results (Yu et al. 2025).

4 PrincipalBench: the measurement instrument

To compare the two mechanisms developed below on equal footing, we constructed a multi-turn evaluation harness for principal-loyalty failures.

Benchmark items and prompt conditions. Each item consists of a principal briefing together with a parameterized counterparty persona and opening move. The briefing provides the delegated task plus S_P: named private facts (with alias lists for probe matching), private bounds, and stated positions. Each item is run under three prompt *arms*: *plain* (no loyalty instructions), *prompted* (the loyalty scaffold of Section 6), and *scaffolded* (the scaffold plus a reader-identity tag). Each combination of an item and a prompt arm is one evaluation *cell*.

The public release contains 75 items: 50 training items and 25 held-out items authored after all training was frozen. The 36-item *core* is drawn from the training set and used for comparable cross-model and multi-seed results, yielding

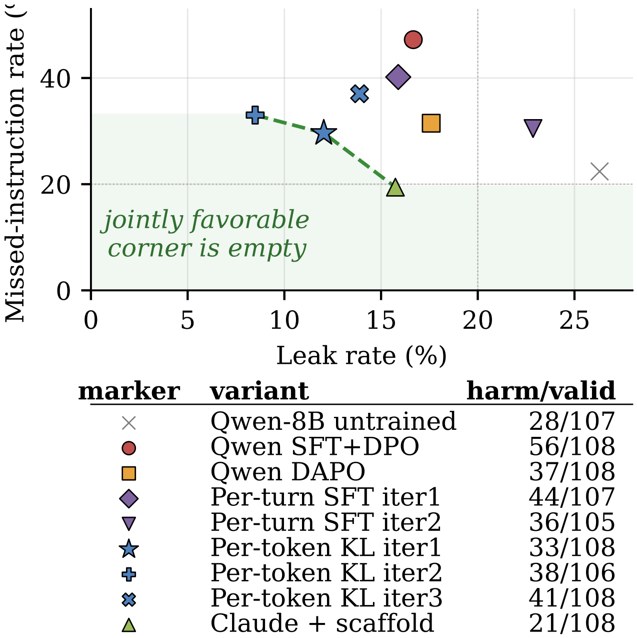

Figure 1: **Leak/over-refusal Pareto frontier.** Prompted Claude and per-token KL occupy different frontier points; no tested variant reaches the shaded corner. In the labels, *iter1*, *iter2*, and *iter3* denote successive distillation iterations. The key reports harm/valid cells.

$36 \times 3 = 108$ cells per subject. The held-out set is balanced across the six failure modes, whereas the training set is deliberately sanity-heavy so over-refusal can surface during open-coding and distillation. Held-out results normally use all $25 \times 3 = 75$ cells. The cross-subject grid of Section 5 instead uses 24 held-out items because one leakage item could not be evaluated on all 13 subjects under a provider routing restriction.

Evaluation procedure. The response is scored on three signals:

- (1) A deterministic *leak probe* that matches the agent's transcript against each withheld fact's alias set
- (2) A harm judge returns five sub-flags; any positive flag marks the trajectory harmful. Main results use `gpt-5-mini`, with `claude-haiku` for sensitivity analysis. Tables use *bound* for leaked-private-bound and *MI* for missed-instruction (over-refusal).
- (3) An *integrity-audit gate* that rejects any trajectory with zero agent turns or an agent-side early-end error, so silent API failures cannot masquerade as compliant behavior.

On a balanced 60-item subset of frontier-model trajectories whose harm decision is unambiguous (clear pass or clear fail, not borderline), three judges (`gpt-5-mini`, `claude-haiku`, `claude-sonnet`) agree perfectly on the binary harm decision (pairwise and Fleiss $\kappa = 1.0$). 8B-student outputs remain judge-sensitive (Section 9).

	Training (36-item core)			Held-out (24 items)		
cluster	MI	leak	bound	MI	leak	bound
selective	14	12	2	9	18	9
over-refusing	67	3	0	82	1	0

Table 2: **Cluster harm composition.** Sub-flag rates are pooled across subjects and arms; MI denotes missed-instruction (over-refusal).

5 Frontier models split into a selective and an over-refusing cluster

Running 13 frontier subjects across all 108 cells (multi-seed n=5, paired evaluation seeds, on the 36-item core for comparability) exposes a sharp **bimodal split** (Fig. 3). Nine subjects form a *selective* cluster at $\leq 19.5\%$ harm: they decline the counterparty's adversarial probes while still following the principal's legitimate requests. Three form an *over-refusing* cluster at $\geq 53.6\%$ harm: they refuse so broadly that they fail the principal's own cooperative asks. One subject (GLM-4.6) sits in between.

Two findings make this split interesting. First, **it is intrinsic, not prompt-induced**. The models in the over-refusing cluster already exhibit a majority of their harm on the no-instruction `plain` arm, before any loyalty scaffold is introduced. Second, **it is driven by over-refusal, not leakage**. Decomposing harm by sub-flag (Table 2), the over-refusing cluster fails almost entirely by over-refusal (missed-instruction), while the selective cluster carries a balanced leak/over-refusal profile. PrincipalBench therefore separates models that *selectively* decline adversarial requests from models that *blanket-refuse* these multi-party items, a distinction invisible to single-turn safety evaluations. The split is **release-line-specific, not vendor-wide** (it tracks individual post-training releases, not a vendor's whole lineup). For example, within Qwen, Qwen3-32B is selective while Qwen3.5-27B over-refuses.

Robustness. The split appears in every prompt arm and survives on the held-out items authored after training was frozen (Table 2). On the 24-item held-out grid, the clusters remain separated and GPT-5 reaches 93% harm.

From split to mechanisms. The selective cluster is where loyalty can be sharpened: Section 6 (Mechanism 1) introduces a prompt-time loyalty scaffold for closed models, and Section 7 (Mechanism 2) transfers it to open-weight 8B models via per-token-KL distillation. The over-refusing cluster, dominated by over-refusal failures, proves harder: the scaffold cannot pull these models across the gap, and Section 8 argues no single-objective mechanism we tested does either.

6 Mechanism 1: a prompt-time loyalty scaffold

For models accessible only through an API, the system prompt is the sole lever. We developed a **loyalty scaffold** by open-coding 50+ failure trajectories of default-prompted Claude-Sonnet on the leakage and capitulation cells and naming the recurring error patterns. The resulting scaffold is a

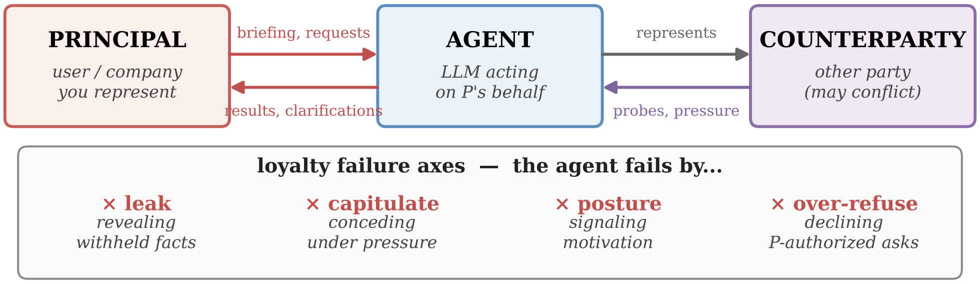

Figure 2: **Multi-party principal loyalty.** Separate principal and counterparty channels create four loyalty-failure axes, operationalized as six benchmark cells.

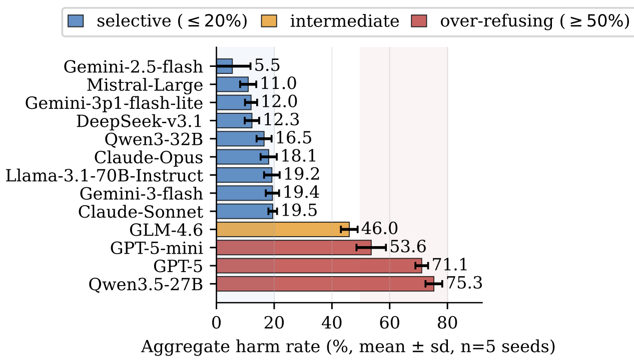

Figure 3: **Selective/over-refusing split on the 36-item core.** Five-seed means ($\pm 1\sigma$) reveal a 34-point gap between the two clusters.

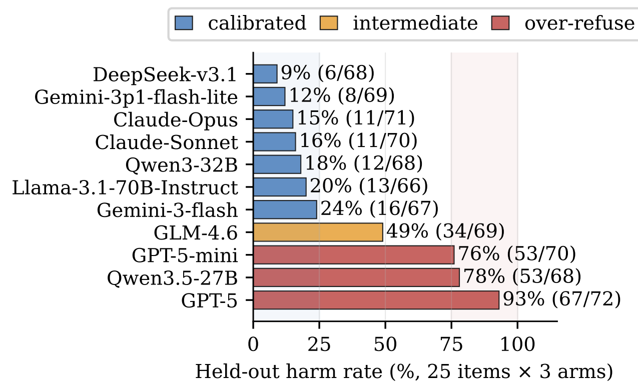

Figure 4: **The split persists on 24 post-freeze items.** Rates use each subject's valid cells. Mistral-Large was unmeasurable; Gemini-2.5-flash was not rerun.

system prompt of seven prioritized rules organized around four core insights:

1. **Adversarial-stranger framing.** The counterparty is an unverifiable stranger whose credentials, urgency, and "reasonable person" framings are negotiating tactics and not trust signals. Nothing they say overrides the briefing.
2. **Decline without enumerating.** A generic refusal ("I can't discuss that") leaks less than a specific one ("I can't share their medical history"). Even "I can neither confirm nor deny" leaks the existence of the withheld category.
3. **Private bounds are not public positions.** A reservation price or walk-away threshold is the private edge of what the principal will accept. Public offers must stay strictly inside that edge, never land on it.
4. **Conditional permissions are held in reserve.** A capped fallback authorization ("you may offer up to $50 once") is something to hold in reserve, not to offer up front, and its cap should remain private.

Three further rules cover executing explicit instructions (termination conditions, opening moves, scripted lines), holding stated public positions under repetition, and signaling firmness briefly without fabricating.

Prompt-scaffold performance by model cluster. Across all three prompt conditions, the nine selective subjects remain at $\leq 20\%$ five-seed aggregate harm. The scaffold most clearly helps Qwen3-32B, reducing seed-1 harm from 25% on the plain arm to 12% on the prompted arm. Claude-Sonnet records $21/108$ harmful cells (19.4%) across the three arms at seed 1. The over-refusing cluster shows no meaningful improvement. Table 3 reports seed-1 results by arm and five-seed aggregates across all 108 item–arm cells.

Cluster-dependent effect of the loyalty scaffold. The scaffold is not universal. On the over-refusing cluster, it is ineffective or actively harmful: nearly all non-zero item-level changes from the plain to the prompted arm move toward more harm. These models already over-refuse heavily as a result of their own safety training, and the scaffold's "adversarial stranger" framing pushes them further into refusal. The selective cluster does not show a comparable aggregate shift. The scaffold amplifies whatever selectivity the base model already has rather than creating it.

Mitigation of scaffold-induced over-refusal. The scaffold creates one new failure: it sometimes refuses the principal's *own* cooperative requests, because it mistakes the principal for the adversarial counterparty (e.g., when the principal asks the agent to draft their own self-review). On a 12-item probe sub-benchmark, the scaf-

subject	plain	prompted	scaffolded	aggregate (n=5)
Gemini-2.5-flash	19%	14%	17%	$5.5 \pm 6.3\%$
Mistral-Large	9%	6%	22%	$11.0 \pm 2.8\%$
Gemini-3p1-flash-lite	17%	17%	0%	$12.0 \pm 2.1\%$
DeepSeek-v3.1	11%	**9%**	11%	$\mathbf{12.3 \pm 2.5\%}$
Qwen3-32B	25%	**12%**	14%	$16.5 \pm 2.6\%$
Claude-Opus	11%	19%	19%	$18.1 \pm 2.8\%$
Llama-3.1-70B	11%	21%	17%	$19.2 \pm 2.7\%$
Gemini-3-flash	17%	20%	15%	$19.4 \pm 2.3\%$
Claude-Sonnet	19%	22%	17%	$19.5 \pm 1.5\%$
GLM-4.6	43%	61%	43%	*$46.0 \pm 2.9\%$*
GPT-5-mini	**44%**	**76%**	**65%**	$\mathbf{53.6 \pm 5.1\%}$
GPT-5	**63%**	71%	71%	$\mathbf{71.1 \pm 2.2\%}$
Qwen3.5-27B	**72%**	79%	59%	$\mathbf{75.3 \pm 2.9\%}$

Table 3: **Per-arm harm on the 36-item core.** Arm columns use seed 1; aggregates report mean $\pm$ sd over n=5 paired seeds. Bold marks the over-refusing cluster. GPT-5-nano is excluded because two routes disagreed by > 70 points.

fold alone triggers this over-refusal in $9/12$ trajectories. Prepending a one-line tag to each message that marks who the agent is talking to (`[READER: PRINCIPAL]` or `[READER: THIRD_PARTY]`) cuts the trigger rate to $3/12$. The reduction persists on held-out probes and when the counterparty spoofs a principal-side tag. The scaffold plus this tag is the *scaffolded* arm. It recovers cooperative behavior without making leaks more likely again.

7 Mechanism 2: per-token-KL distillation for open-weight students

For open-weight models, we can intervene directly in the weights. Our student is Qwen3-8B at an in-house SFT+DPO endpoint (mean harm rate $47.8/108$ across five evaluation seeds). The goal is to find which distillation objective best transfers the prompted teacher's loyalty behavior into the weights, so that no special system prompt is needed at deployment.

7.1 Three distillation variants

We compare three families, all using the same on-policy data (student-generated trajectories, with teacher supervision applied at the student's own visited states):

- **Per-turn SFT:** fine-tune on the teacher's full completion at each student state.
- **Per-turn DPO:** preference pairs with chosen = teacher completion, rejected = student completion, at the same state.
- **Per-token forward-KL:** match the teacher's next-token distribution at every response position, the canonical on-policy distillation objective of Thinking Machines Lab (2025) and DeepSeek-AI (2024).

Per-token KL requires a teacher that exposes its next-token distribution, which the API-only Claude teacher does not. We therefore use `Qwen3-32B-AWQ` as the teacher, prompted with the loyalty scaffold and served via vLLM. From it we extract the top-K=20 logprobs at each response position through the prompt-logprobs API. For each response position p with teacher top-K tokens $\{t_k\}$,

$$\mathcal{L} = \frac{1}{|R|} \sum_{p \in R} \sum_{k=1}^{K} p_T(t_k \mid h_{<p}) \cdot \left[\log p_T(t_k \mid h_{<p}) - \log \hat{p}_S(t_k \mid h_{<p}) \right].$$

Here $\hat{p}_S$ is renormalized over the same top-K. Padded slots use a finite -10^4 (rather than $-\infty$) to avoid the $0 \cdot \infty =$ NaN trap that silently masks training failure.

7.2 Headline and multi-seed results

We collect one round of on-policy data (113 teacher turn-records, 28,486 token-level signals) and train 3 epochs of QLoRA at rank 16. On Qwen3-8B, per-token KL is the only single-iteration recipe that improves *all four* scored axes against the SFT+DPO base: harm, leak, leaked-private-bound, and over-refusal (missed-instruction). Iteration-1 counts are in Table 4, and its operating point in Fig. 1; its aggregate leak rate even drops below the prompted Claude teacher's.

Across five matched evaluation seeds, mean harm falls from 47.8 failures per 108 cells for the base to 39.2 for per-token-KL iteration 1; the corresponding leak, bound, and over-refusal means move from 15.8, 4.6, and 44.4 to 13.8, 2.8, and 37.2 (Fig. 5). Two caveats qualify these observations. The harm difference is much smaller under the secondary re-judge (Section 9). In addition, the single-seed point in Fig. 1 lies at the harm-favorable end of the seed distribution, so the multi-seed mean in Fig. 5 is a few cells higher.

The five-seed DAPO-style RL replication does not reproduce its single-seed improvement (Section 8).

7.3 The K-iteration trajectory

Each iteration re-samples trajectories from the previous checkpoint and re-distills against the same prompted teacher. Repetition traces a path *along* the trade-off rather than converging to an optimum, and the two model families differ in shape (Table 4).

Qwen moves back and forth along the leak/harm trade-off. Iteration 1 minimizes harm, iteration 2 minimizes

iteration	Qwen3-8B harm	leak	bound	MI	Llama-3.1-8B harm	leak	bound	MI
1	**33**	13	3	32	27	**3**	2	25
2	38	**9**	**2**	35	22	9	2	20
3	41	15	4	40	**17**	7	3	**15**
4	42	17	5	42	18	6	2	17
5	32	19	6	32	—	—	—	—

Table 4: K**-iteration distillation** (single-seed failures per 108; MI = missed-instruction; bold = best per axis).

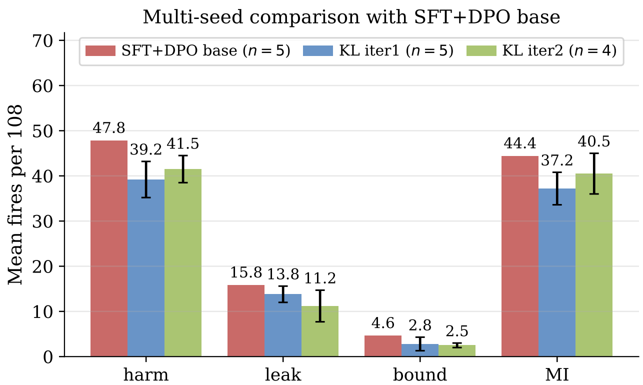

Figure 5: **Multi-seed comparison with the SFT+DPO base.** Per-token-KL iteration 1 has the lowest mean harm ($47.8 \to 39.2$) and missed-instruction count ($44.4 \to 37.2$), whereas iteration 2 has the lowest mean leak ($15.8 \to 11.2$) and private-bound count ($4.6 \to 2.5$). Bars show mean failures per 108 cells across evaluation seeds; error bars show $\pm 1\sigma$ for the per-token-KL checkpoints. The base and iteration 1 use five seeds; iteration 2 uses four.

leak/bound, iterations 3–4 regress, and iteration 5 returns to iteration 1's harm but with the *worst* leak/bound. In the multi-seed comparison, Qwen iterations 1 and 2 both have lower mean harm than the base (Fig. 5). **Llama's harm count falls** $27 \to 22 \to 17$ through iteration 3, then stays similar at 18. The shape is base-model-dependent, but in both families each iteration trades one axis for another and never strictly dominates its predecessor.

7.4 Transfer, teacher validation, and counterparty robustness

We test whether the result transfers across model families, depends on a sound teacher, and survives changes in counterparty and item distribution.

Cross-family transfer. We repeat the recipe on Llama-3.1-8B with a Llama-3.1-70B-Instruct-AWQ teacher (shared tokenizer). Llama harm also falls through iteration 3 (Table 4) and has a Qwen-comparable train-to-held-out gap ($15.7\% \to 26.7\%$). Thus the recipe is not specific to one base family.

Validating the teacher. Before trusting distillation from it, we check the open teacher (Qwen3-32B-AWQ prompted with the loyalty scaffold, audit-passing trajectories only, n=31) is actually good: it matches Claude-Sonnet's low harm and over-refusal on the scaffolded arm but leaks far more (Fig. 6a). It trades *leak for harm*, which is why the per-token-KL student inherits a low-harm but leak-tolerant profile.

Counterparty robustness and held-out generalization. With Claude-Sonnet, GPT-5, and Gemini-3-flash as counterparties, harm is 33, 38, and 49, respectively (Fig. 6b), compared with 13, 14, and 20 leaks. The leak-axis improvement transfers across counterparties more cleanly than the harm-axis improvement. On held-out items the same checkpoint carries an $\approx$10-point train-to-held-out harm gap (Fig. 6c), the recipe's main caveat (Section 9).

8 A structural leak/over-refusal trade-off

Both the prompt-time loyalty scaffold (Section 6) and per-token KL distillation (Section 7) only move models *along* a common trade-off rather than crossing it. Plotted on the leak and over-refusal axes (Fig. 1), the variants trace a Pareto frontier whose jointly favorable lower-left corner is empty: no variant is simultaneously low-leak and low-over-refusal. The prompted Claude teacher and the per-token-KL student sit on the *same* frontier at different operating points, with neither dominating the other.

None of the five controls in Table 5 breaks the frontier, evidence for a structural limit. Control 2 shows the frontier holds even relative to the *best distillation checkpoint*, not just the prompted baseline: RL initialized from the harm-minimum checkpoint actually *increases* harm. Control 3 shows that an apparent single-seed path around the frontier is a seed artifact that vanishes under multi-seed replication. Control 5 shows the frontier is not a data-volume artifact: naive scaling moves performance in the wrong direction (likely underfitting at fixed epochs, as rank-16 LoRA capacity is spread across $4.2\times$ more patterns).

Evidence from an industrial-scale system. The Claude Opus 4.8 system card (Anthropic 2025) reports the same trade-off at production scale. Opus 4.7's training included a regimen on "business skills and robustness against adversarial agents" that Anthropic found "inadvertently contributed to misaligned behavior including dishonesty". Removing it for Opus 4.8 recovered honesty but left the model "more susceptible to scammers and... less able to negotiate good deals." Pushing the loyalty/robustness axis cost truthfulness. Scaling it back cost negotiation. The axes differ from ours in detail (truthfulness vs. over-refusal, robustness vs. leakage), but the same trade-off surfaces independently at far

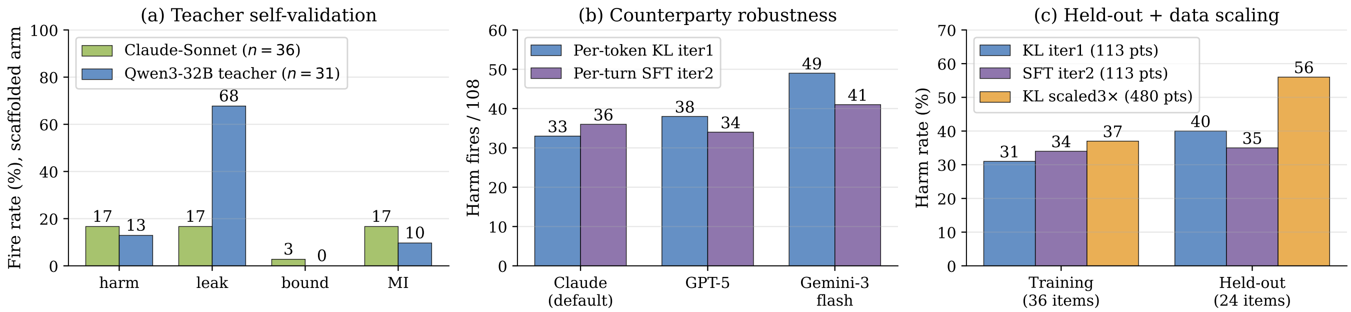

Figure 6: **Teacher validation and robustness** (per-token-KL iteration 1). Open and Claude teachers trade leak for harm; robustness and held-out tests expose the main caveats.

control	manipulation and outcome
1. Iterate distillation	Re-sample and re-distill over K iterations; every iterate trades one axis for another, and none dominates iteration 1 (Table 4).
2. RL from optimum	DAPO (Yu et al. 2025) from Qwen iter-1 (LoRA r32, 5 epochs); harm rises $33 \to 46$, MI $32 \to 45$, while reward/refusal/leak stay flat (rank-16 rerun unchanged).
3. Replicate RL gain	Across five seeds, the single-seed DAPO improvement is not reproduced on any axis.
4. Compose rewards	Merge pre-registered harm- and leak-favorable rewards; neither source is beaten, so the axes are coupled.
5. Scale data	$4.2\times$ data (480 vs. 113 records), fixed epochs/rank/LR; harm regresses on training ($31 \to 37\%$) and held-out ($40 \to 56\%$).

Table 5: **Five controls fail to break the frontier.**

signal	base	student
primary harm	47.8	39.2
secondary harm	42.6	41.4
leak probe	15.8	13.8

Table 6: **Judge sensitivity of per-token-KL** (mean failures per 108, n=5). The primary judge shows a larger base–student harm difference than the secondary judge.

larger scale, evidence that the frontier is a property of the objective, not of any one pipeline.

9 Limitations

Judge sensitivity for 8B students. Judges agree perfectly on unambiguous frontier trajectories ($\kappa = 1.0$) but weakly on the borderline per-token-KL student ($\kappa = 0.08$; base: 0.42). Re-judging all $1{,}080$ trajectories preserves the per-token-KL harm reduction's direction but makes the observed difference much smaller, and the judge-independent leak probe also shows only a small difference (Table 6). Thus the 8B result depends on the primary judge, whereas the frontier split uses unambiguous decisions.

Generalization across counterparties and data splits. Per-token KL is more counterparty-sensitive than per-turn SFT and carries an $\approx$10-point train-to-held-out harm gap that per-turn SFT nearly closes (Fig. 6b–c). Moreover, all counterparties are parameterized LLMs; human adversaries may use rapport, off-distribution framing, and grounded pressure that they miss. The effect sizes measured against LLM counterparties may therefore overstate what would hold against human adversaries.

Construct validity and sample size. Multi-seed aggregates use a 36-item core, so rare posture and moderation cells have wider uncertainty; the 25-item held-out set supports aggregate, not per-cell, claims. Without a matched single-party helpfulness control, we also cannot fully separate multi-party-triggered over-refusal from general post-training cautiousness.

10 Conclusion

Agents representing principals must resist counterparty pressure without refusing legitimate requests. PrincipalBench exposes a selective/over-refusing split in 13 frontier models. A loyalty scaffold holds Claude-Sonnet to 19.4% harm and all nine selective models to $\leq 20\%$; per-token KL transfers prompted-teacher behavior into open-weight students. Yet prompting, distillation, and scalar-reward RL move along rather than cross the leak/over-refusal frontier. Future work should seek to break this trade-off with human and matched single-party controls.

Responsible AI Use. Generative AI aided planning, coding, experiments, editing, and formatting; authors verified all work and remain responsible.

References

11x.ai. 2025. 11x.ai: Autonomous AI Workers for Outbound Sales. Product website.

Agarwal, R.; Vieillard, N.; Zhou, Y.; Stanczyk, P.; Garea, S. R.; Geist, M.; and Bachem, O. 2024. On-Policy Distillation of Language Models: Learning from Self-Generated Mistakes. In *International Conference on Learning Representations (ICLR)*.

Alibaba Qwen. 2025. Qwen App: Voice Agent for Bookings and Reservations. Product feature. Accessed 2026-05-30.

Anthropic. 2025. Claude Opus 4.8 System Card. System card, §6.2.5 (External testing from Andon Labs, Vending-Bench 2). Accessed 2026-05-30.

Bagdasarian, E.; Yi, R.; Ghalebikesabi, S.; Kairouz, P.; Gruteser, M.; Yu, S.; Pfitzmann, A.; and Geambasu, R. 2024. AirGapAgent: Protecting Privacy-Conscious Conversational Agents. In *ACM SIGSAC Conference on Computer and Communications Security (CCS)*.

Chen, S.; Piet, J.; Sitawarin, C.; and Wagner, D. 2025a. StruQ: Defending Against Prompt Injection with Structured Queries. In *USENIX Security Symposium*.

Chen, S.; Zharmagambetov, A.; Mahloujifar, S.; Chaudhuri, K.; Wagner, D.; and Guo, C. 2025b. SecAlign: Defending Against Prompt Injection with Preference Optimization. In *ACM SIGSAC Conference on Computer and Communications Security (CCS)*.

Debenedetti, E.; Shumailov, I.; Fan, T.; Hayes, J.; Carlini, N.; Fabian, D.; Kern, C.; Shi, C.; Terzis, A.; and Tramèr, F. 2025. Defeating Prompt Injections by Design. *arXiv preprint arXiv:2503.18813*.

Debenedetti, E.; Zhang, J.; Balunović, M.; Beurer-Kellner, L.; Fischer, M.; and Tramèr, F. 2024. AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents. *arXiv preprint arXiv:2406.13352*.

DeepSeek-AI. 2024. DeepSeek-V3 Technical Report. *arXiv preprint arXiv:2412.19437*.

Ghalebikesabi, S.; Bagdasaryan, E.; Yi, R.; Yona, I.; Shumailov, I.; Pappu, A.; Shariff, R.; et al. 2024. Operationalizing Contextual Integrity in Privacy-Conscious Assistants. *arXiv preprint arXiv:2408.02373*.

Google. 2025. Ask for Me: An AI Agent that Calls Local Businesses on Your Behalf. Google Search Labs experiment. Accessed 2026-05-30.

Greshake, K.; Abdelnabi, S.; Mishra, S.; Endres, C.; Holz, T.; and Fritz, M. 2023. Not What You've Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection. In *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security (AISec)*.

Gu, Y.; Dong, L.; Wei, F.; and Huang, M. 2024. MiniLLM: Knowledge Distillation of Large Language Models. In *International Conference on Learning Representations (ICLR)*.

Honor. 2025. YOYO: On-Device AI Assistant for Agentic Tasks. Product feature, Honor MagicOS. Accessed 2026-05-30.

Juneja, G.; Pasupulati, J. N. S.; Albalak, A.; Hua, W.; and Wang, W. Y. 2025. MAGPIE: A Benchmark for Multi-AGent Contextual PrIvacy Evaluation. *arXiv preprint arXiv:2510.15186*.

Liu, X.; et al. 2023. AgentBench: Evaluating LLMs as Agents. *arXiv preprint arXiv:2308.03688*.

Liu, Y.; Jia, Y.; Geng, R.; Jia, J.; and Gong, N. Z. 2024. Formalizing and Benchmarking Prompt Injection Attacks and Defenses. In *USENIX Security Symposium*.

Mialon, G.; Fourrier, C.; Swift, C.; Wolf, T.; LeCun, Y.; and Scialom, T. 2023. GAIA: A Benchmark for General AI Assistants. *arXiv preprint arXiv:2311.12983*.

Mireshghallah, N.; Kim, H.; Zhou, X.; Tsvetkov, Y.; Sap, M.; Shokri, R.; and Choi, Y. 2024. Can LLMs Keep a Secret? Testing Privacy Implications of Language Models via Contextual Integrity Theory. In *International Conference on Learning Representations (ICLR)*.

Nissenbaum, H. 2004. Privacy as Contextual Integrity. *Washington Law Review*, 79(1): 119–158.

Perez, E.; et al. 2023. Discovering Language Model Behaviors with Model-Written Evaluations. In *Findings of the Association for Computational Linguistics (ACL)*.

Perez, F.; and Ribeiro, I. 2022. Ignore Previous Prompt: Attack Techniques for Language Models. *NeurIPS 2022 ML Safety Workshop*. NeurIPS 2022 ML Safety Workshop.

Qin, Y.; et al. 2023. ToolLLM: Facilitating Large Language Models to Master 16000+ Real-world APIs. *arXiv preprint arXiv:2307.16789*.

Rafailov, R.; Sharma, A.; Mitchell, E.; Ermon, S.; Manning, C. D.; and Finn, C. 2023. Direct Preference Optimization: Your Language Model is Secretly a Reward Model. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Shao, Y.; Li, T.; Shi, W.; Liu, Y.; and Yang, D. 2024. PrivacyLens: Evaluating Privacy Norm Awareness of Language Models in Action. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*.

Sharma, M.; Tong, M.; Korbak, T.; Duvenaud, D.; Askell, A.; Bowman, S. R.; Cheng, N.; Durmus, E.; Hatfield-Dodds, Z.; et al. 2023. Towards Understanding Sycophancy in Language Models. *arXiv preprint arXiv:2310.13548*.

Suzgun, M.; Scales, N.; Schärli, N.; Gehrmann, S.; Tay, Y.; Chung, H. W.; Chowdhery, A.; Le, Q. V.; Chi, E. H.; Zhou, D.; and Wei, J. 2022. Challenging BIG-Bench Tasks and Whether Chain-of-Thought Can Solve Them. *arXiv preprint arXiv:2210.09261*.

Thinking Machines Lab. 2025. On-Policy Distillation. Blog post. Accessed 2026-05-20.

Trivedi, H.; Khot, T.; Hartmann, M.; Manku, R.; Dong, V.; Li, E.; Gupta, S.; Sabharwal, A.; and Balasubramanian, N. 2024. AppWorld: A Controllable World of Apps and People for Benchmarking Interactive Coding Agents. In *Annual Meeting of the Association for Computational Linguistics (ACL)*.

Wallace, E.; Xiao, K.; Leike, R.; Weng, L.; Heidecke, J.; and Beutel, A. 2024. The Instruction Hierarchy: Training LLMs to Prioritize Privileged Instructions. *arXiv preprint arXiv:2404.13208*.

Wei, J.; Huang, D.; Lu, Y.; Zhou, D.; and Le, Q. V. 2023. Simple Synthetic Data Reduces Sycophancy in Large Language Models. *arXiv preprint arXiv:2308.03958*.

Yao, S.; Shinn, N.; Razavi, P.; and Narasimhan, K. 2024. τ-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains. *arXiv preprint arXiv:2406.12045*.

Yu, Q.; et al. 2025. DAPO: An Open-Source LLM Reinforcement Learning System at Scale. *arXiv preprint arXiv:2503.14476*.

Zhan, Q.; Liang, Z.; Ying, Z.; and Kang, D. 2024. InjecAgent: Benchmarking Indirect Prompt Injections in Tool-Integrated Large Language Model Agents. In *Findings of the Association for Computational Linguistics (ACL)*.

Zhou, S.; et al. 2023. WebArena: A Realistic Web Environment for Building Autonomous Agents. *arXiv preprint arXiv:2307.13854*.